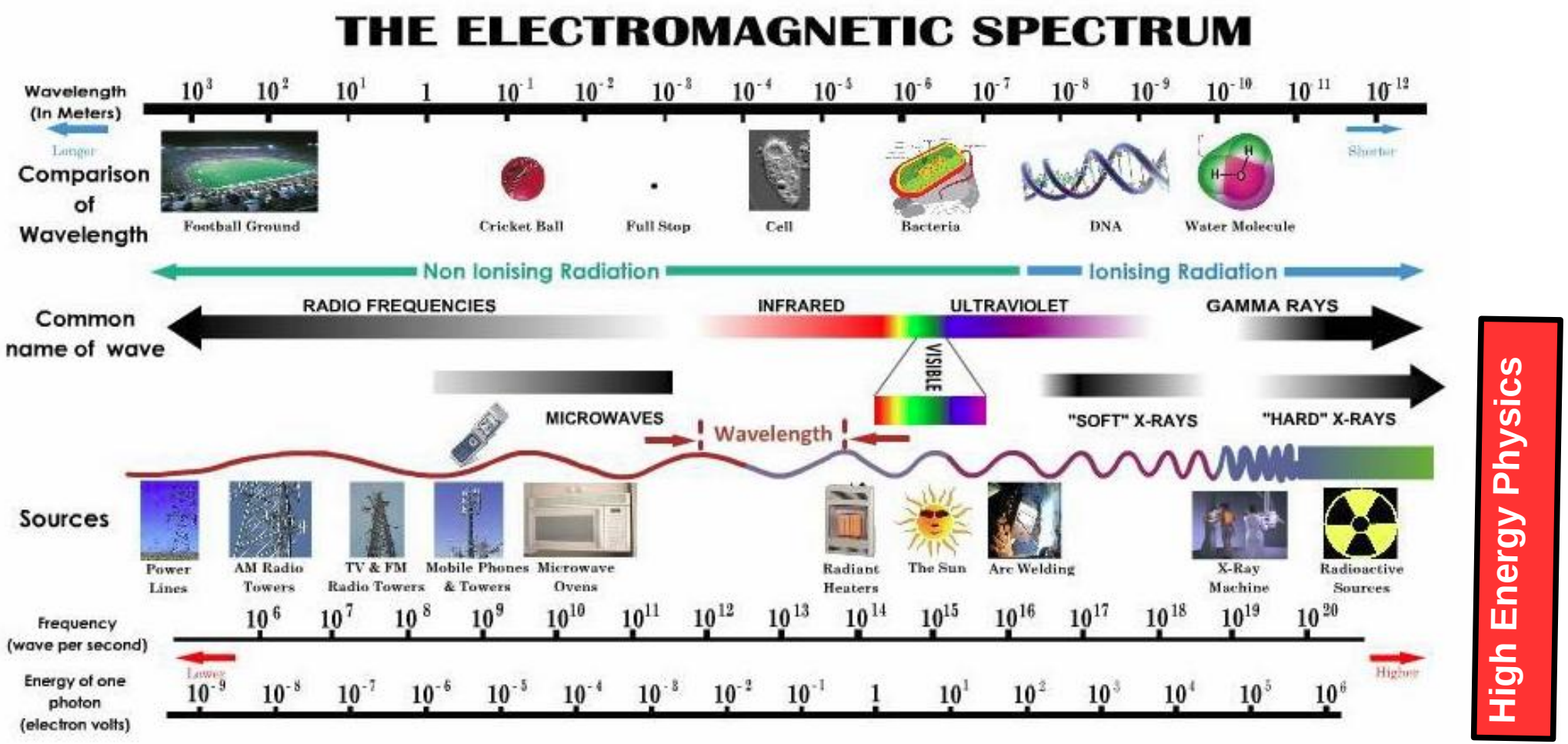
Light : Source of Knowledge

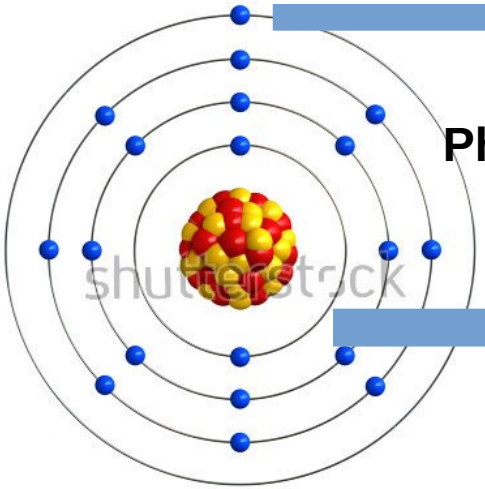


High Energy Physics

$$E=mc^2$$

Atom

Excitation Energy

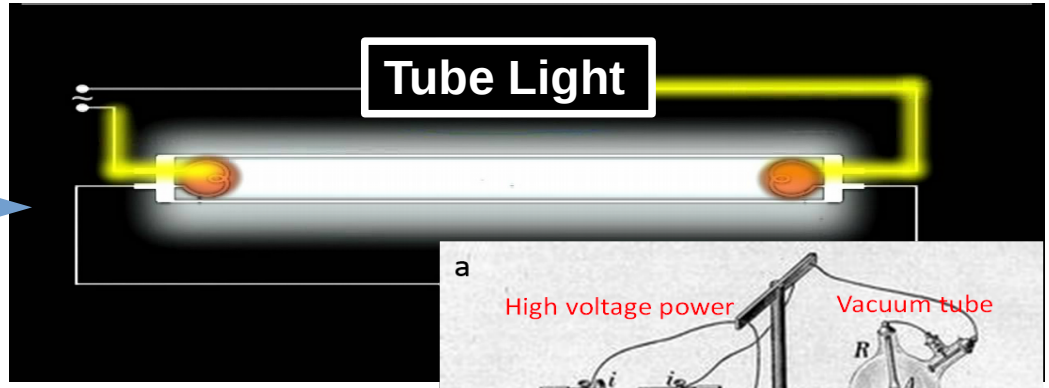


> eV

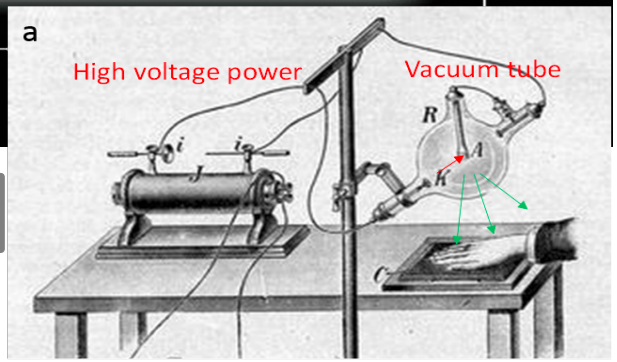
Photon (visible range)

> KeV

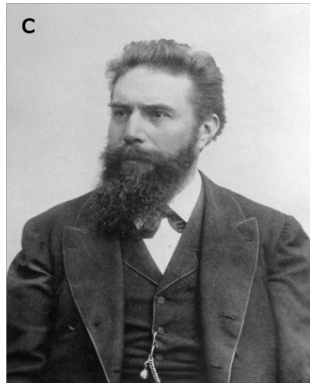
X-Ray

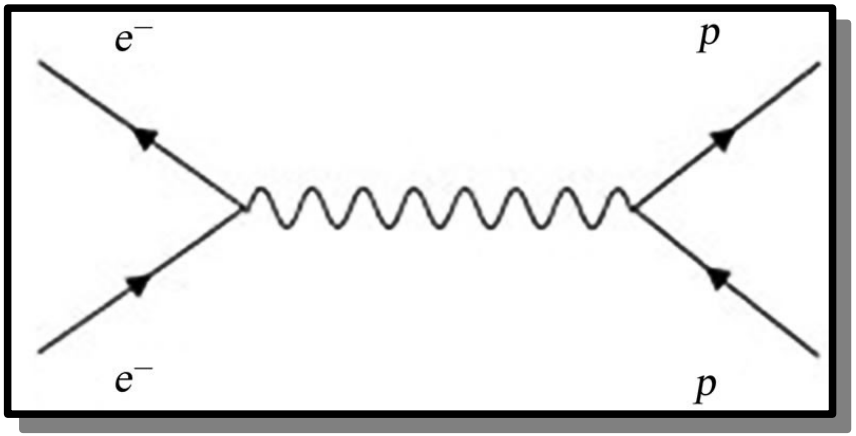


Spectra of QED



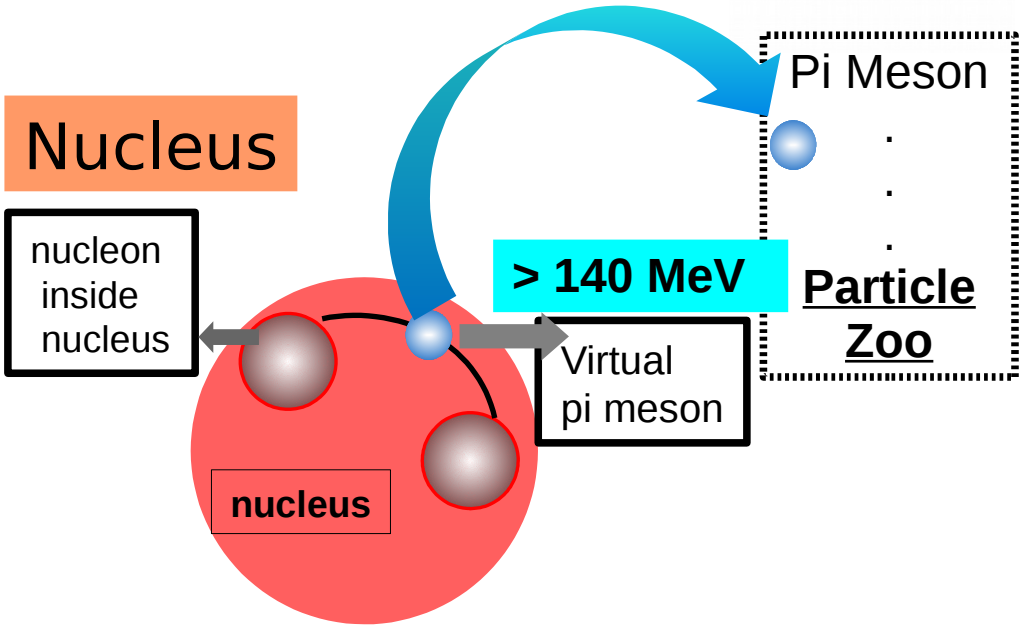


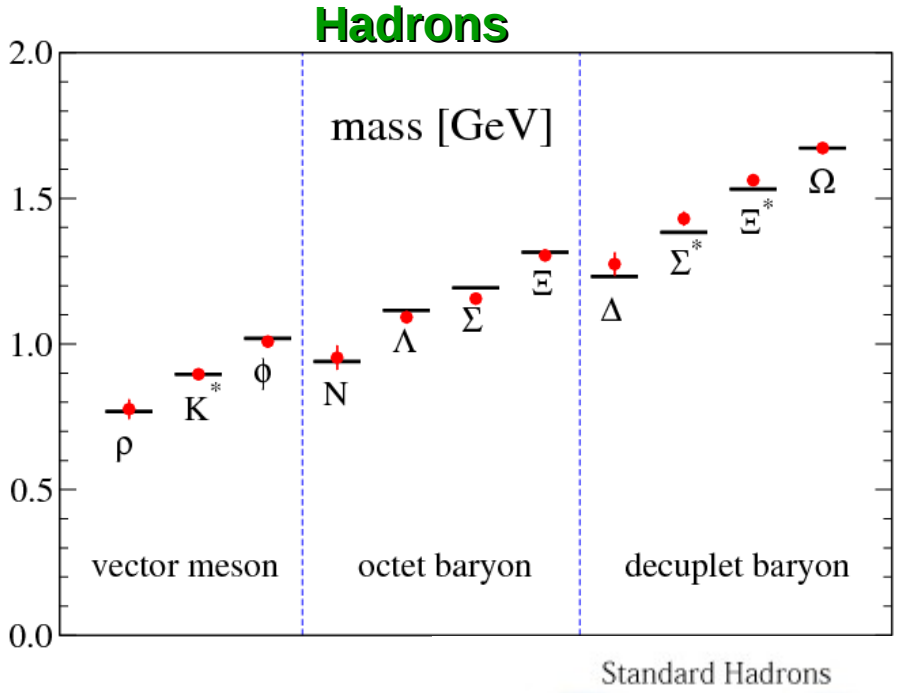




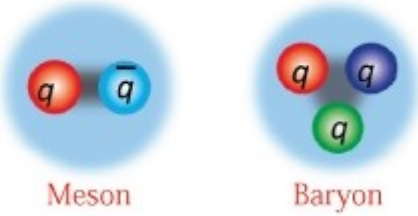


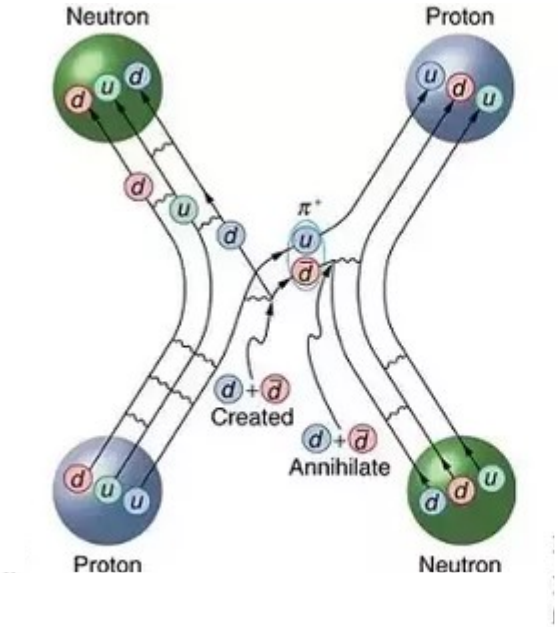
$$E=mc^2$$





Strong Interaction





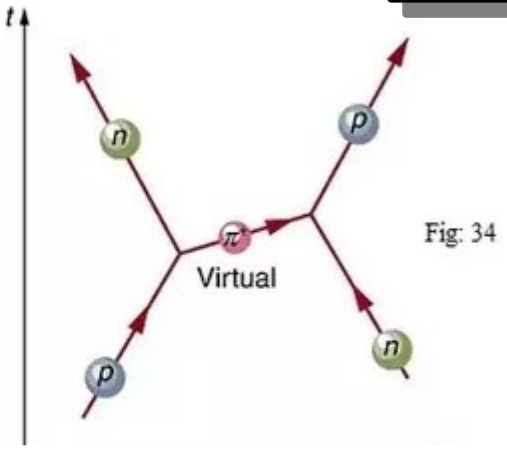
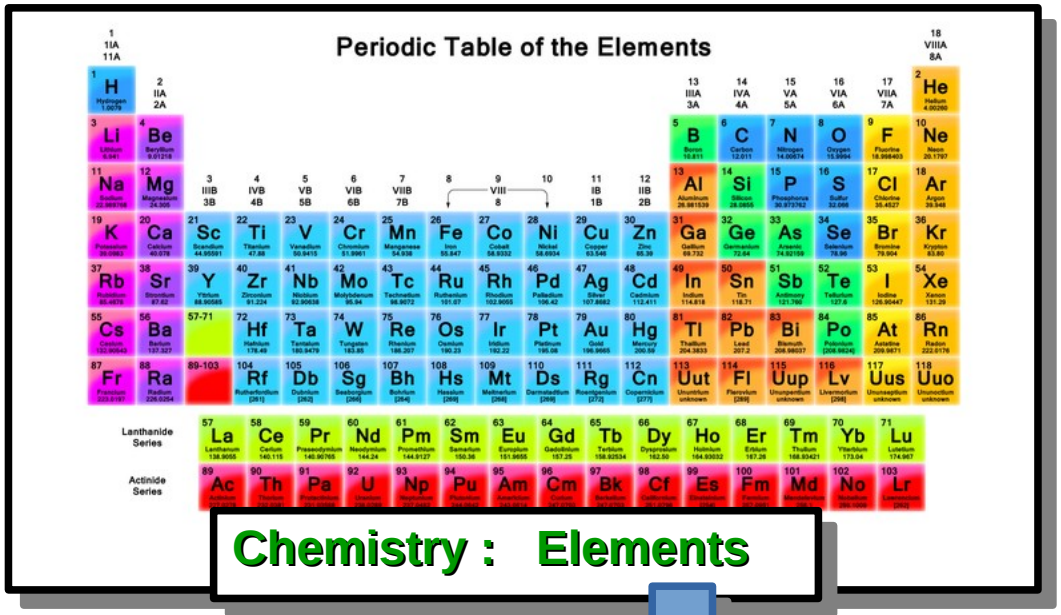


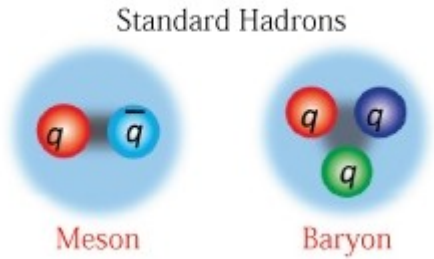
Fig: 34

2.4 MeV ⅔ **u** up	1.27 GeV ⅔ **c** charm	171.2 GeV ⅔ **t** top
4.8 MeV -⅓ **d** down	104 MeV -⅓ **s** strange	4.2 GeV -⅓ **b** bottom



Compounds

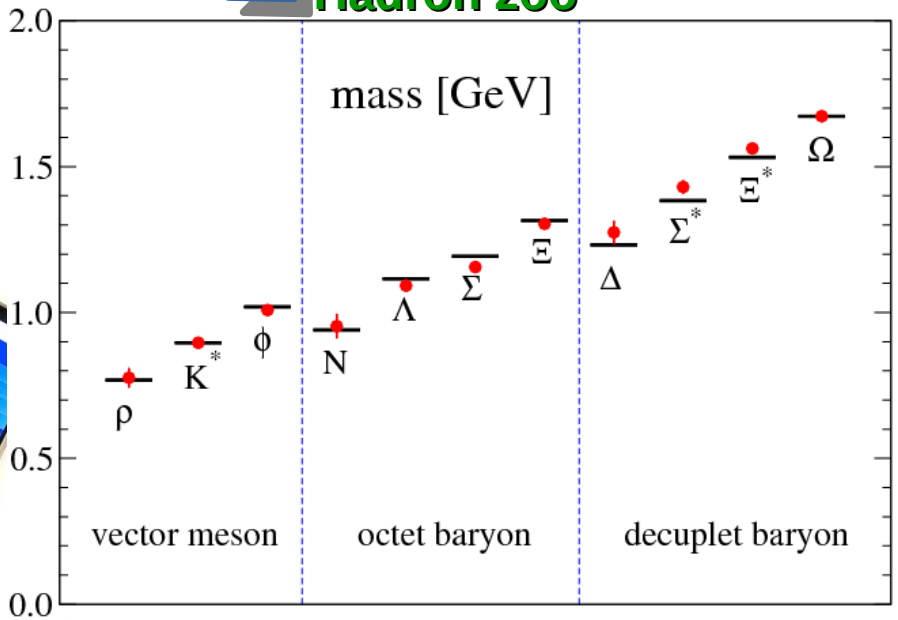
Quark Chemistry



2.4 MeV ⅔ u up	1.27 GeV ⅔ c charm	171.2 GeV ⅔ t top
4.8 MeV -⅓ d down	104 MeV -⅓ s strange	4.2 GeV -⅓ b bottom

Hadron zoo





~ few MeV



VECC, Kolkata, India



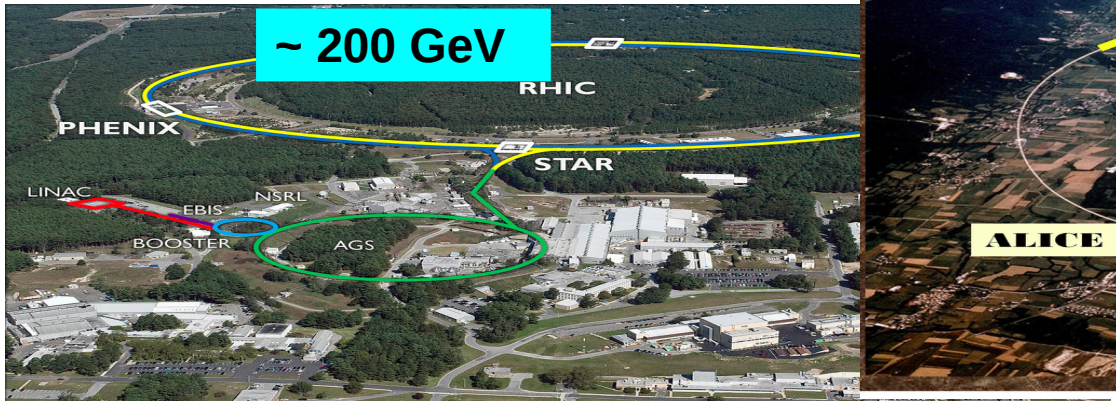
IUAC, Delhi, India

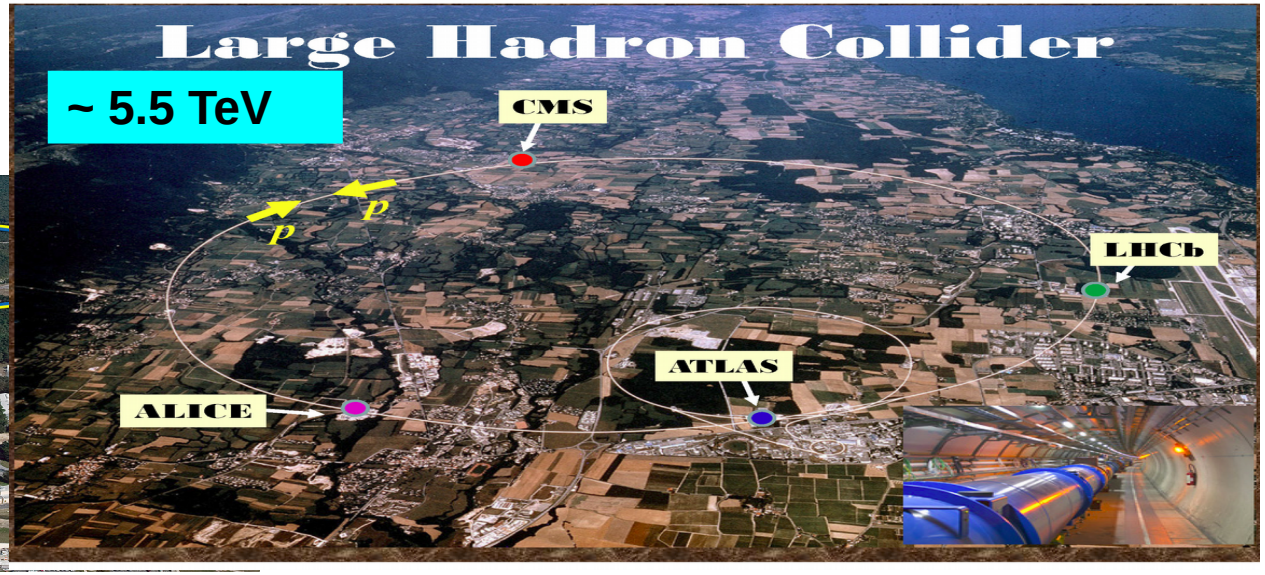
National Status

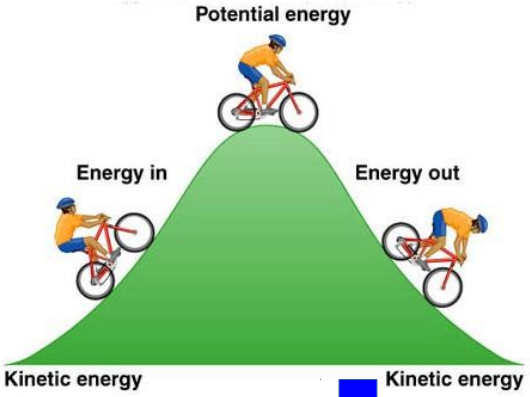
Accelerators

International Status

RHIC, BNL, USA



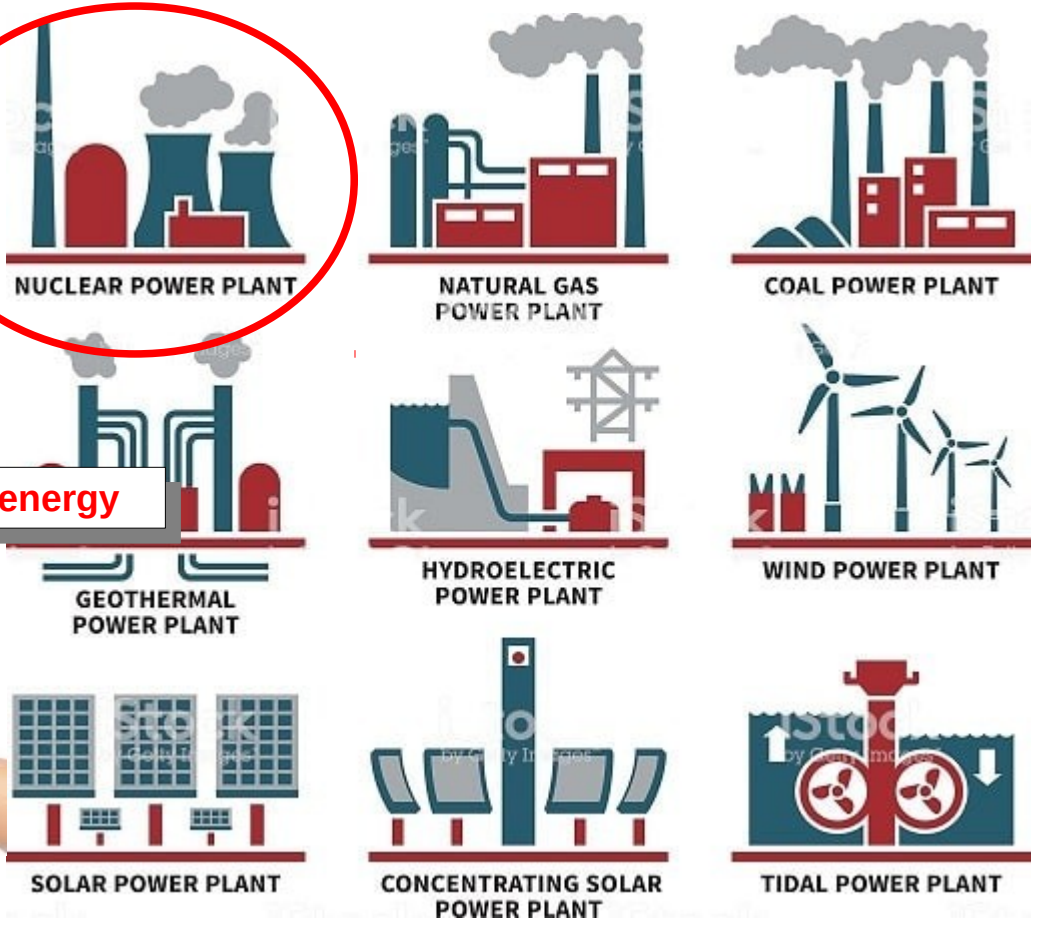




Fortunately Nature don't allow you extract rest mass energy..... Nuclear physics unfold a possibility (but still not entire rest mass)..... ☺

$$E^2 = \boldsymbol{p}^2 c^2 + m_0^2 c^4 ,$$

Rest mass (Hidden) energy



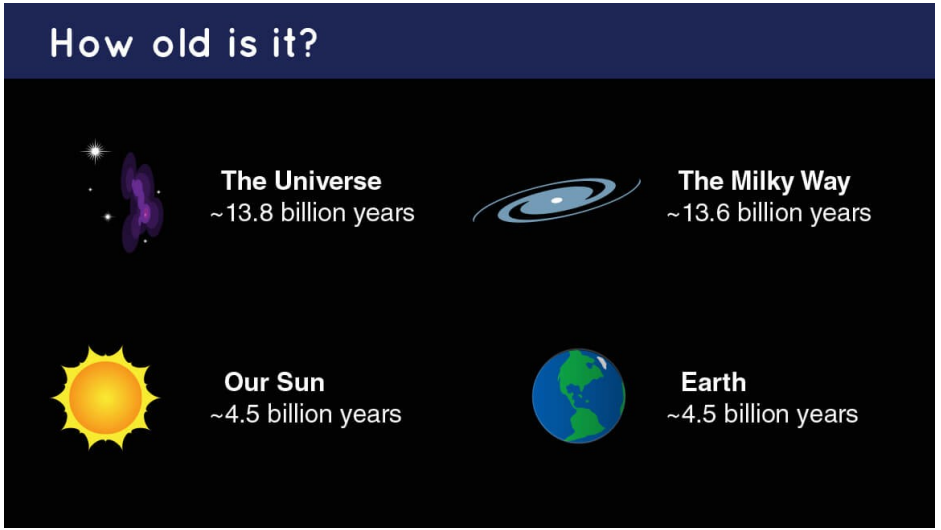


Byproduct of basic science become presenta day's entertainment to daily life .....



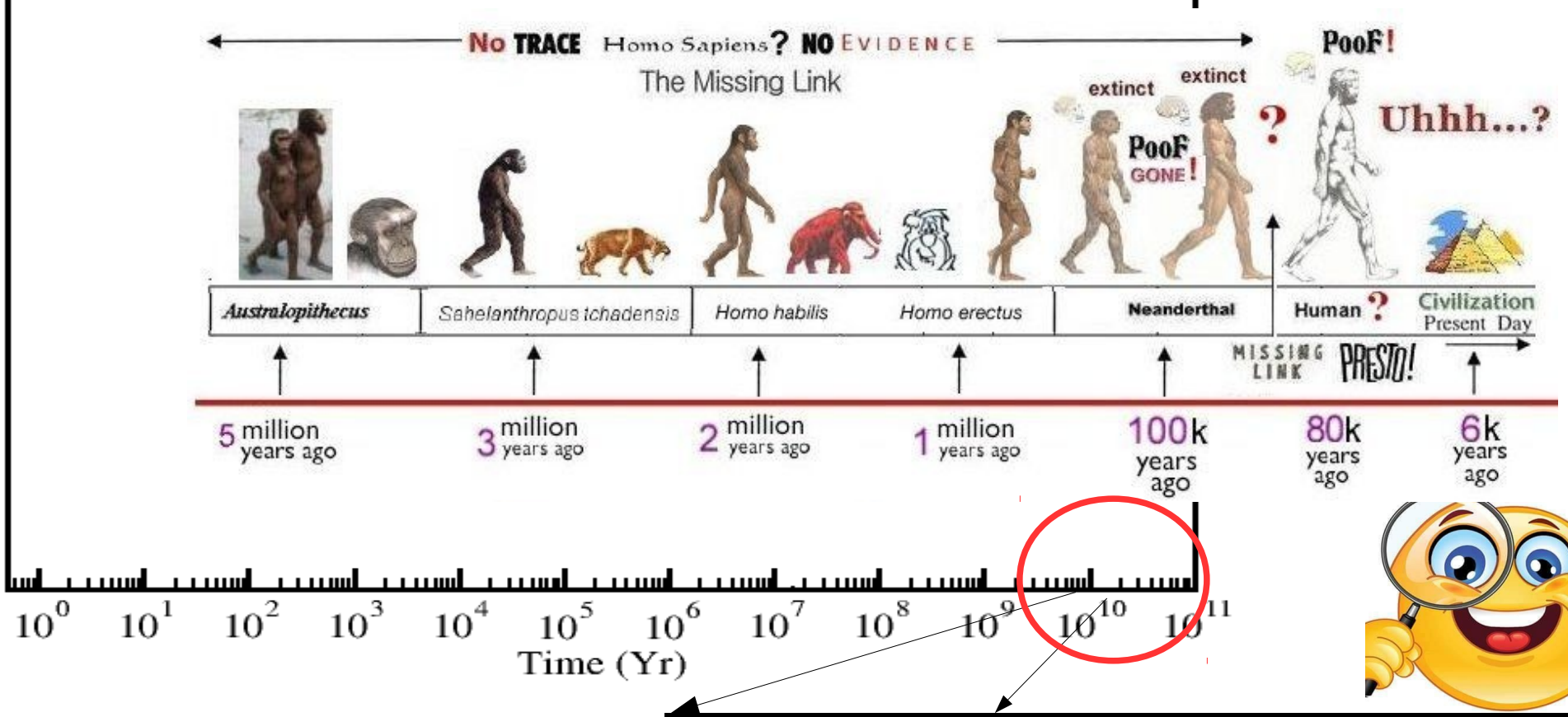


Trying to orient our brain about entire time axis of our unverse life time

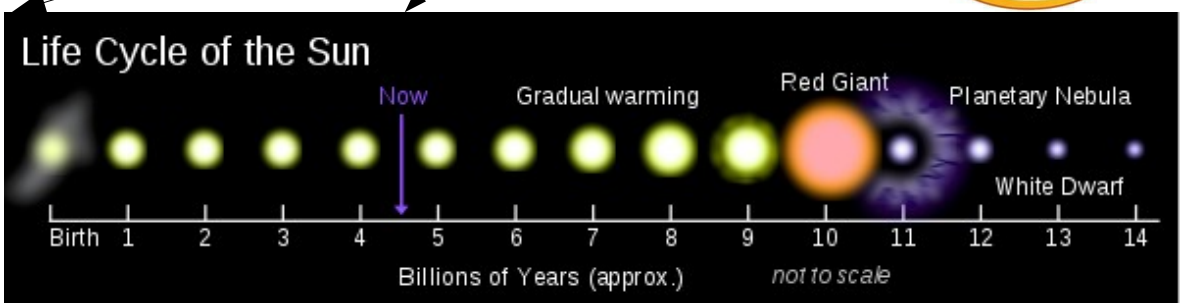


Number in words	Number in figures	Number in standard form	Number written as a decimal
One thousand	1,000	10^3	
Ten thousand	10,000	10^4	0.01 million
One hundred thousand	100,000	10^5	0.1 million
One million	1,000,000	10^6	
Ten million	10,000,000	10^7	0.01 billion
One hundred million	100,000,000	10^8	0.1 billion
One billion	1,000,000,000	10^9	
Ten billion	10,000,000,000	10^{10}	0.01 trillion
One hundred billion	100,000,000,000	10^{11}	0.1 trillion
One trillion	1,000,000,000,000	10^{12}	
One quadrillion	1,000,000,000,000,000	10^{15}	



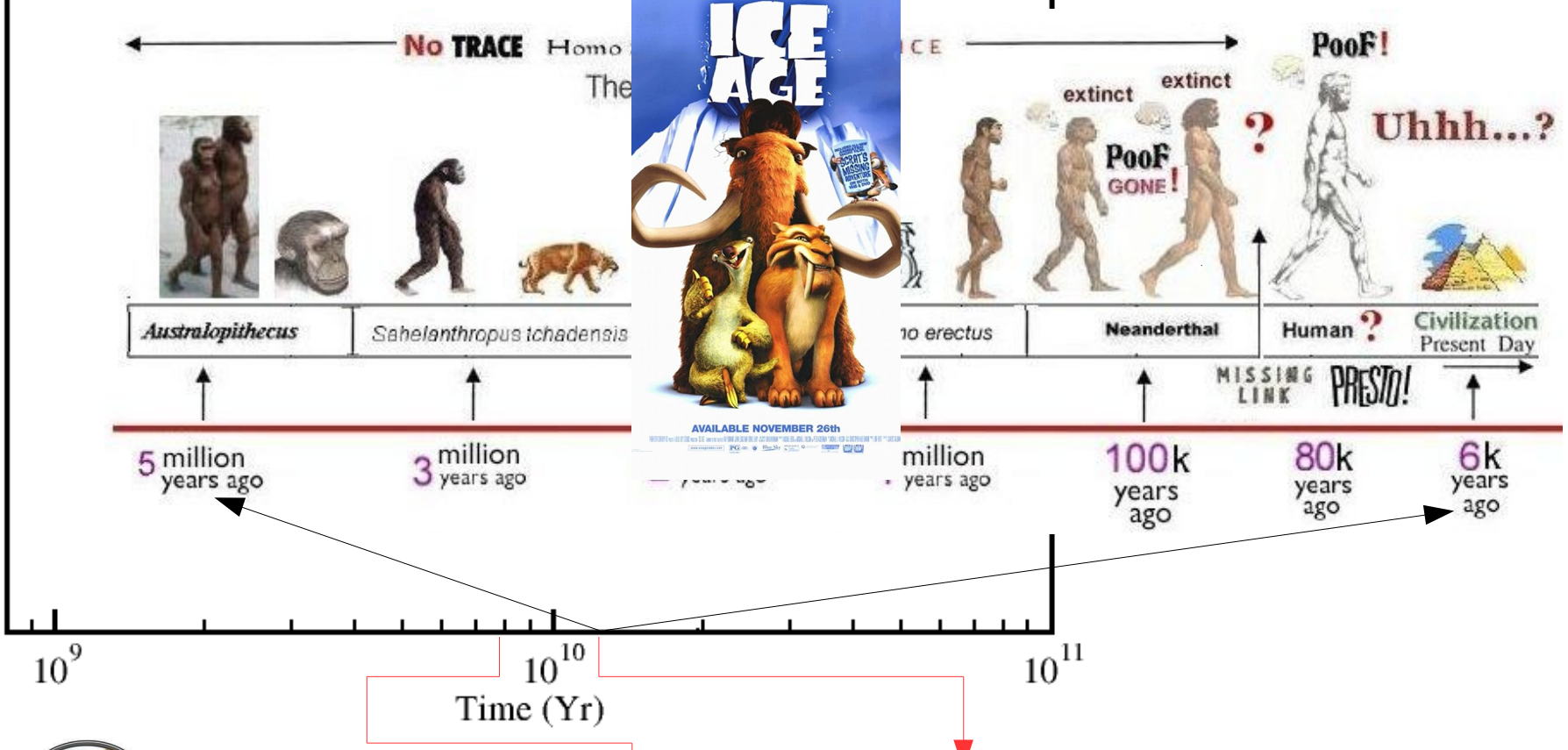




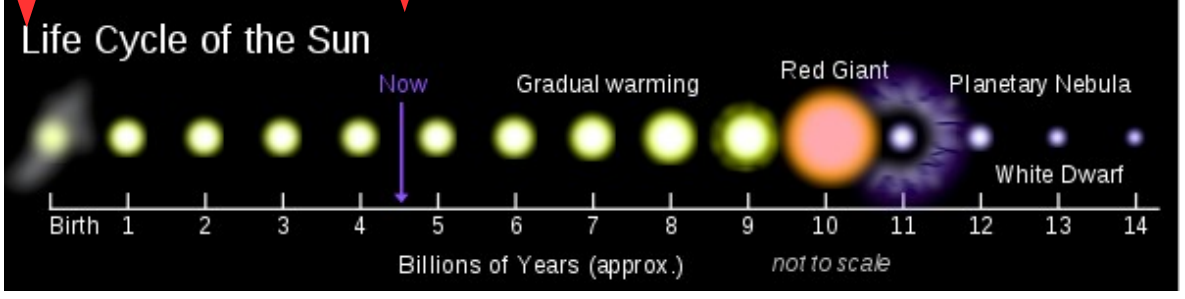


YouTube

https://www.youtube.com/watch?v=JgPwTA_o90s


No TRACE
Homo
The
ICE AGE
AVAILABLE NOVEMBER 26th
PooF!
extinct
extinct
PooF
GONE!
?
Uhhh...?
Australopithecus
Sahelanthropus tchadensis
Neanderthal
Human ?
Civilization
Present Day
MISSING LINK
PRESTO!
5 million years ago
3 million years ago
100k years ago
80k years ago
6k years ago
10^9
10^{10}
10^{11}
Time (Yr)




Life Cycle of the Sun
Now
Gradual warming
Red Giant
Planetary Nebula
White Dwarf
Birth 1 2 3 4 5 6 7 8 9 10 11 12 13 14
Billions of Years (approx.)
not to scale

Wikipedia Link: https://en.wikipedia.org/wiki/Cosmic_Calendar

